

(How much) should I care about my grandchildren's future?

ECON-4210

Economics of Climate Change

Class 9

Prof. Casey Wichman

Today's class

1. Questions?

2. Attendance

3. Housekeeping

1. Problem sets and exams are not yet graded – hopefully by Monday.
2. Problem set 2 will be assigned in early/mid October.
3. Exam 2 on Monday, 10/19.

4. Today

A. Discounting in the long run

5. For next class...

Time is money...

Would you rather have \$600 now or \$1000 in 10 years?

Time is money...

Would you rather have \$600 now or \$1000 in 10 years?

- PV of \$1000 in 10 years is:
 - $\$1000 / (1 + r)^t = \$1000 / (1 + 0.05)^{10} = \613.91
 - (Assuming 5% interest rates)
- Generally, people are impatient and would prefer money now rather than later.
 - The discount rate (which might be the interest rate on their savings account) governs how impatient they are.
- How do we think about how *society* trades off time for money?

How much would we be willing to pay today to avoid destruction of Florida in 200 years?

- NASA has just learned that in September 2226, Florida will be annihilated by an asteroid.
- We can stop this today with a missile strike (don't worry about logistics...), but it costs \$10M today.
- Assume the value of land and capital in Florida is \$2 trillion in 2226.
- How much should we be willing to pay to save Florida?



*Not really, this is just a thought experiment!

Value of FL	Discount Rate	PV of saving FL?
\$2T	7%	
\$2T	3%	
\$2T	1%	

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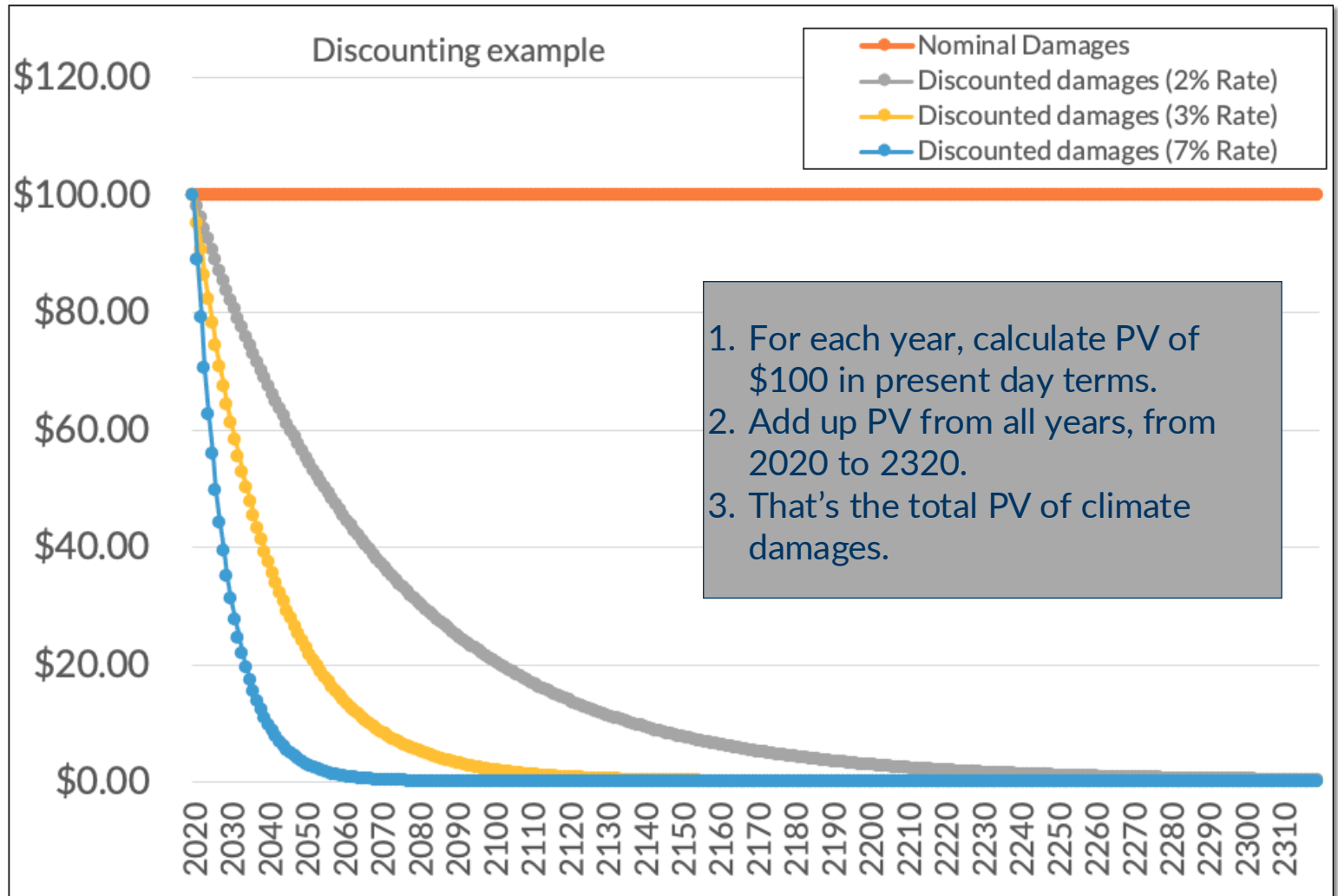
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Impact of discount rate on “long-run” damages

- Imagine a *stream* of climate damages of \$100/year for 300 years.
 - That is, every year, climate change generates \$100 of damage.
 - Highly simplified, but isolates impact of discount rates.

PV of total damages	
Nominal damages	\$30,000.00
PV Damages (2% Rate)	\$5,086.59
PV Damages (3% Rate)	\$2,076.28
PV Damages (7% Rate)	\$905.53



Alternative approaches to discounting

- **Descriptive (or opportunity-cost) approach** to discounting...
 - We should invest resources according real rates of return
 - You could take \$50M and build a wind farm.
 - Or, you could invest \$50M in financial markets and earn 7% (or more) returns per year.
 - Therefore, discount rates should be determined by the *opportunity cost of capital*.

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 - ▶ Therefore, discount rates should be determined by the *opportunity cost of capital*.
- **Prescriptive (or normative) approach** to discounting...
 - ▶ It is unethical to “discount” the welfare of future generations, so we should use a very low discount rate (e.g., 1% or lower)
 - ▶ Important to distinguish discount rates on *consumption* vs. discount rates on *welfare* of future generations
 - ▶ Future generations will (very likely) be wealthier than us, so we might want to value consumption in our relatively poor generation more heavily than consumption in the future (*even if we value future generations' welfare equally!*)

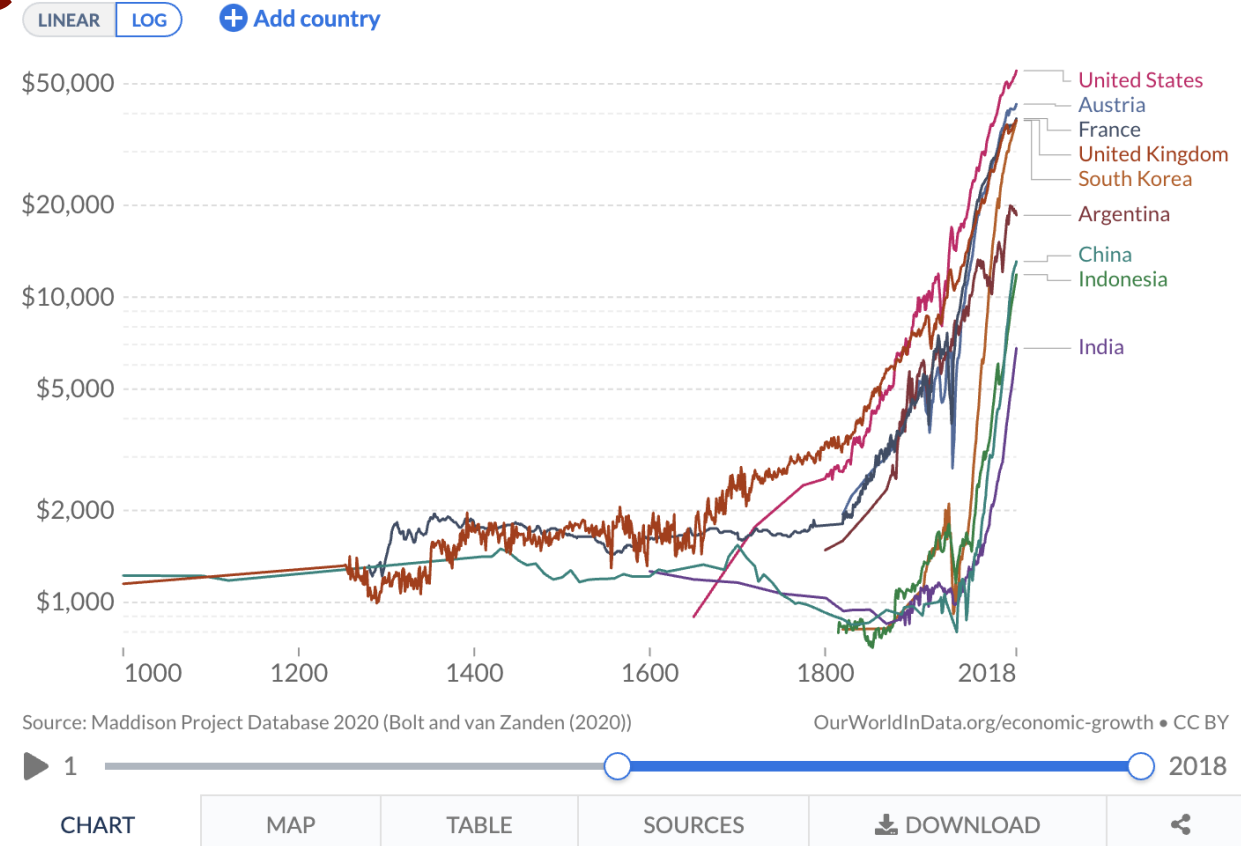
Income growth over time

- Future generations will (very likely) be wealthier than us.
- Generally, we value an additional \$1 paid to Elon Musk less than an additional \$1 paid to a college student.
 - This idea is based on diminishing marginal utility. This is the same reason why demand curves slope downwards.
- In the long run, our generation is a college student and our great grandchildren are Elon Musk.
- This is one (of several) justifications for the prescriptive approach to discounting over long time horizons.
 - Note: even if we value the welfare of future generations equally, this “*aversion to inequality*” would still justify discounting.

GDP per capita, 1000 to 2018

GDP per capita adjusted for price changes over time (inflation) and price differences between countries – it is measured in international-\$ in 2011 prices.

Our World
in Data



Note: Y-axis in log

The Social Discount Rate

via the Ramsey equation (a prescriptive approach to discounting...)

$$r = \delta + \eta g$$

- r is the **social discount rate**.
- δ is the **pure rate of time preference**
 - Governs how consumers discount *utility* over time. Higher δ raises marginal benefit of consuming now.
 - If you want to weight future generations' welfare the same as today's generation, you'd set $\delta = 0$.
- η is the **elasticity of the marginal utility of consumption**
 - Governs how much we value an additional dollar as society grows wealthier. As η increases, the more we dislike inequality, which means we value today's consumption more strongly.
 - a.k.a, "Coefficient of relative risk aversion" or "intergenerational inequality aversion"
- g is the **expected per capita growth rate of consumption**
- δ and η are ethical parameters (and, generally, subject to much disagreement)

The Social Discount Rate

$$r = \delta + \eta g$$

$$\delta + \eta \times g = r$$

TABLE 6-1 Values and Implied Social Discount Rates in Selected Studies

Author	Rate of Pure Time Preference (in %)	Risk/Inequality Aversion	Anticipated Growth Rate (in %)	Implied Social Discount Rate (in %)
Cline (1992)	0.0	1.5	1.0	1.5
IPCC (1996)	0.0	1.5-2.0	1.6-8	2.4-16
Arrow (1999)	0.0	2	2.0	4.0
UK: Green Book (HM Treasury, 2003)	1.5	1	2.0	3.5 ^a
U.S. Office of Management and Budget (2003) ^b				3-7
France: Rapport Lebègue (2005)	0.0	2	2.0	4.0 ^a
Stern (2007)	0.1	1	1.3	1.40
Arrow (2007)		2-3		
Dasgupta (2007)	0.1	2-4		
Weitzman (2007)	2.0	2	2.0	6.0
Nordhaus (2008)	1.0	2	2.0	5.0

NOTES: The table shows the calibration of the discount rate based on the Ramsey rule; see text for discussion.

^aDecreasing with the time horizon.

^bOMB uses a descriptive approach.

SOURCE: Adapted from Intergovernmental Panel on Climate Change (2014c, Table 3-2).

Source: National Academies of Sciences, "Valuing Climate Damages: Updating Estimation of the Social Cost of Carbon Dioxide", 2017.
<https://www.nap.edu/catalog/24651/valuing-climate-damages-updating-estimation-of-the-social-cost-of>

What discount rates do we use?

OMB's Circular A-4 dictates what discount rates to use for BCA...

- Pre-2023 guidance:
 - ▶ **7% discount rate**
 - Opportunity cost of capital
 - Average before-tax rate of return to private capital in the US (somewhat outdated, though)
 - *Hard to justify for climate change because **climate damages are measured in forgone consumption**.*
 - ▶ **3% discount rate**
 - ...if effects of regulation affect consumption
 - More closely reflects “social rate of time preference” (or how society trades-off consumption today vs. tomorrow.)
- In 2023, Circular A-4 was updated to reflect new data/science
 - ▶ **2% as a central discount rate**
 - Real rate of return on “risk-free” long-term government debt.
 - Better reflects updated data on market interest rates
 - ▶ **A lower or “declining” rate for longer-run effects:**
 - Lower rate justified by “Ramsey-like” prescriptive approach to discounting
 - Declining rate justified by uncertainty in the discount rate

Discounting long-run effects in Circular A-4

Arguments for a prescriptive approach

Some believe that it is ethically impermissible to discount the utility of future generations.¹⁶⁸ That is, government should treat all generations equally. Even under an approach that does not discount the utility of future generations, it is often appropriate to discount long-term consumption benefits and costs—although at a lower rate than the near-term effects more likely to fall on a single generation—if there is an expectation that future generations will be wealthier and thus will value a marginal dollar of benefits or costs by less than those alive today, or if there is a non-zero probability of sufficiently catastrophic risks. To account for these special ethical considerations, an extensive literature uses a “prescriptive” approach to long-term discounting, determining the appropriate degree of weight that society should place on the welfare of future generations.

Discounting long-run effects in Circular A-4

Arguments for a lower/declining approach

A distinct reason for discounting the benefits and costs accruing to future generations at a lower rate is uncertainty about the appropriate value of the discount rate.¹⁶⁹ Private market rates provide a reasonably reliable reference for determining the rate at which society is willing to trade consumption over time within a few decades, but for extremely long time periods no comparable private rates exist. Because future changes in the social rate of time preference are uncertain but correlated over time, the certainty-equivalent discount rate will have a declining schedule.¹⁷⁰ The appropriate discount rate declines because it is the average of the cumulative discount factors, not an average of the discount rates, that matters.¹⁷¹

Uncertainty and discounting

- It's difficult to predict the future.
- Interest rates, and thus opportunity costs of investments, could be different in the future.
- Recall $r = \delta + \eta g$.
 - g is the growth rate of consumption per capita, which is uncertain because income and population themselves are uncertain, therefore r is uncertain.
 - **Uncertainty in the discount rate provides justification to use a lower (or declining) discount rate.**

Uncertainty and discounting

PV of \$1,000 delivered in year t (from Arrow et al., 2013)

- Imagine that in an uncertain world, a 1% discount rate and a 7% discount rate are equally likely.
 - Does that mean we can just use a 4% discount rate (i.e., an average of 1% and 7%)?

Years	PV at 1%	PV at 4%	PV at 7%	Expected PV: equally likely chance of 1% or 7%
1	\$990.05	\$960.79	\$932.39	\$961.22
10	\$904.84	\$670.32	\$496.59	\$700.71
50	\$606.53	\$135.34	\$30.20	\$318.36
100	\$367.88	\$18.32	\$0.91	\$184.40
200	\$135.34	\$0.34	\$0.00	\$67.67
300	\$49.79	\$0.01	\$0.00	\$24.89

Uncertainty and discounting

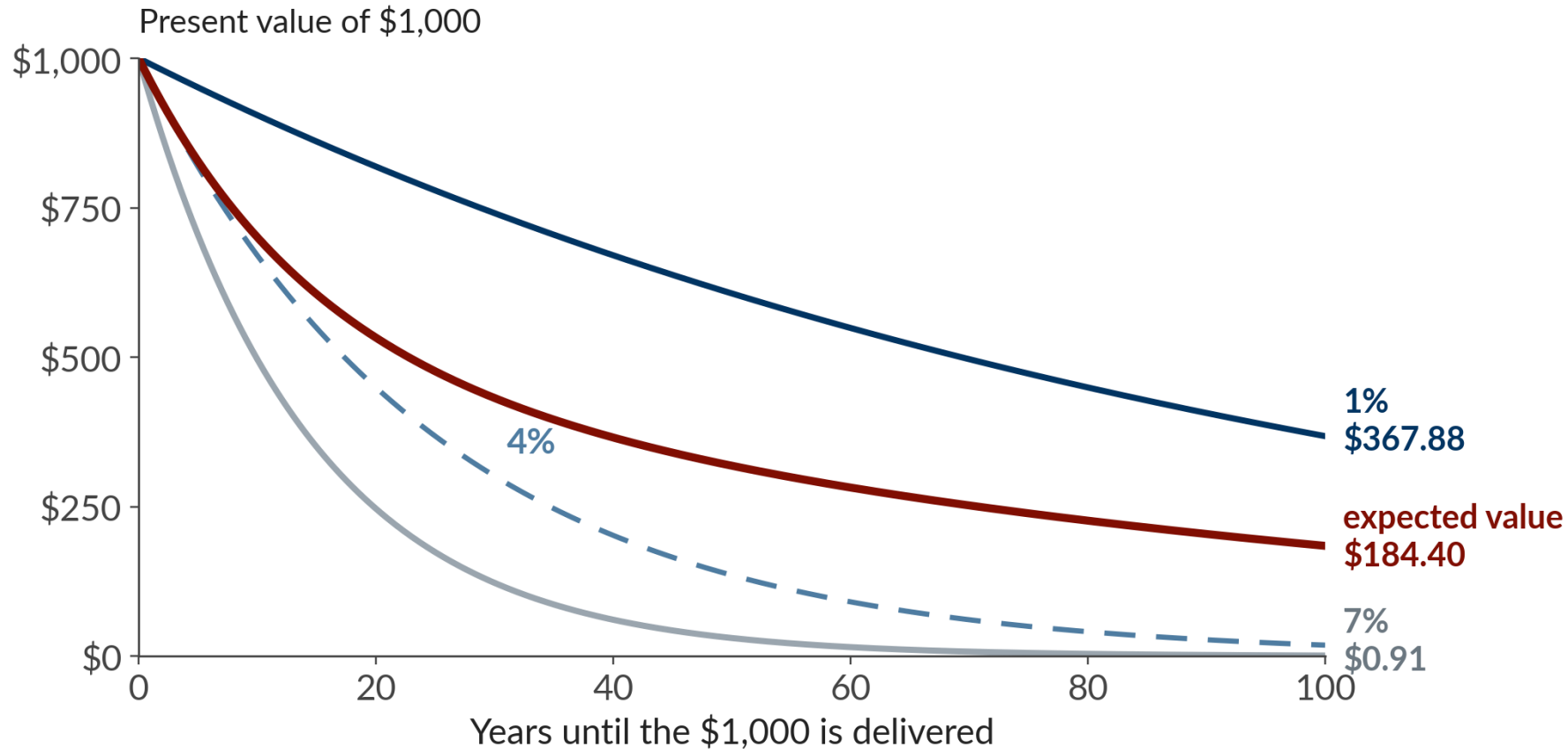
- At 1% the present value is \$367.88. At 7%, 91 cents.
- The average of the two is \$184.40.
- Discounting at 4%, the midpoint rate, gives \$18.32.
- The expected present value is ten times the value at the midpoint rate.

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Uncertainty and discounting

The PV of a low rate (1%) adds far more than the PV of a high rate (7%).

- Raising the rate from 4% to 7% removes \$17.41 of present value.
- Lowering the rate from 4% to 1% adds \$349.56, twenty times as much.



Uncertainty and discounting

We can write the expected present value as a single discount rate.

- ▶ A constant rate of **1.7%** produces a PV of \$184.40 for \$1,000 delivered in 100 years.
- ▶ Equal chances of 1% and 7% are worth the same as a certain 1.7% rate, well below the 4% midpoint.
 - That 1.7% is the "*certainty-equivalent discount rate*."
 - NB: The Arrow et al. (2013) article uses a different, but more complicated, approach to generating the CE discount rate, but the logic is the same.
- ▶ **Main takeaway:**
 - Uncertainty about the discount rate **lowers** the rate that should be used.

Uncertainty and discounting

The longer the time horizon, the lower the certainty-equivalent (CE) rate.

- After 10 years, the CE rate is 3.6%, barely below the 4% midpoint.
- After 300 years the equivalent rate is 1.2%.
- Climate damages arrive at long horizons, so a climate benefit-cost analysis warrants a lower rate than a ten-year regulation.

Years	PV at 1%	PV at 4%	PV at 7%	Expected PV: equally likely chance of 1% or 7%	Constant CE rate that produces Expected PV
1	\$990.05	\$960.79	\$932.39	\$961.22	4.0%
10	\$904.84	\$670.32	\$496.59	\$700.71	3.6%
50	\$606.53	\$135.34	\$30.20	\$318.36	2.3%
100	\$367.88	\$18.32	\$0.91	\$184.40	1.7%
200	\$135.34	\$0.34	\$0.00	\$67.67	1.3%
300	\$49.79	\$0.01	\$0.00	\$24.89	1.2%

Clean Power Plan—Benefits & Costs

- Recall the CPP regulatory impact analysis we discussed before the exam...
- Look at the impact the choice of discount rate for climate benefits can have.

Table ES-9. Monetized Benefits, Compliance Costs, and Net Benefits Under the Rate-based Illustrative Plan Approach (billions of 2011\$) ^a						
Rate-Based Approach						
	2020		2025		2030	
Climate Benefits ^b						
5% discount rate	\$0.80		\$3.1		\$6.4	
3% discount rate	\$2.8		\$10		\$20	
2.5% discount rate	\$4.1		\$15		\$29	
95th percentile at 3% discount rate	\$8.2		\$31		\$61	
Air Quality Co-benefits Discount Rate						
	3%	7%	3%	7%	3%	7%
Air Quality Health Co-benefits ^c	\$0.70 to \$1.8	\$0.64 to \$1.7	\$7.4 to \$18	\$6.7 to \$16	\$14 to \$34	\$13 to \$31
Compliance Costs ^d	\$2.5		\$1.0		\$8.4	
Net Benefits ^e	\$1.0 to \$2.1	\$1.0 to \$2.0	\$17 to \$27	\$16 to \$25	\$26 to \$45	\$25 to \$43
Non-Monetized Benefits	Non-monetized climate benefits					
	Reductions in exposure to ambient NO ₂ and SO ₂					
	Reductions in mercury deposition					
	Ecosystem benefits associated with reductions in emissions of NO _x , SO ₂ , PM, and mercury					
	Visibility impairment					

Discounting takeaways

- Higher discount rates lead to much lower values of damages that occur far in the future.
- Choice of a discount rate embeds ethical choices about how to value future generations and inequality in consumption.
 - Ramsey equation ($r = \delta + \eta g$) separates discount rate into “impatience” and “intergenerational inequality aversion” times economic growth
- Discount rates can be pivotal in deciding whether to move forward with a project.
- Uncertainty about the discount rate lowers the rate that should be used, and lowers it further the longer the time horizon.
 - The UK uses a declining discount rate for intergenerational costs/benefits.

For next class...

www.caseyjwichman.com/econ4210#week6